Project Proposal Report

SkillTwin — AI-Powered Student Skill Intelligence

Only for course Teacher						
		Needs Improvement	Developing	Sufficient	Above Average	Total Mark
Allocate mark & Percentage		25%	50%	75%	100%	20
Level of Understanding	3					
Team Role Fulfillment/Content	7					
Cooperation with Others	5					
Delivery/Q/A	5					
Total obtained mark						
Comments						

Semester: Fall 2026

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Submission Date: 03/10/2026

SkillTwin AI

Project Proposal

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1. Introduction

SkillTwin is a web platform for universities. It reads the work that students submit (lab solution and GitHub projects) and finds the skills shown in that work. It then compares these skills with the skills needed for the student's target job, and helps the student close the gap using AI-generated learning and practical tasks. The university gets summary data about skills, so it can plan workshops and courses.

1.1 Background Overview

- **Domain:** higher education and career development.
- **Current practice.** Universities keep grades and transcripts. Teachers give lab tasks and check them by reading PDFs and code. Students read job advertisements on their own to decide what to learn. Workshops and short courses are usually planned from teacher opinion or surveys.
- **Use of AI today.** AI is used in education for plagiarism checking, automatic code testing, course suggestions, resume writing and chatbots. These tools work separately. None of them turns student work into a standard skill profile that connects to job needs and university planning.

1.2 Problem Statement

- **Problem:** students and teachers cannot see which practical skills a student really shows, or what is missing for a target job.
- **Gaps:** grades hide skills, manual review is slow, and the same skill has many names.
- **Affected:** Students are the main users affected by this problem. They may find it difficult to identify their current skills and manually compare them with the skills required for different job roles.
- **Limits of existing systems.** Plagiarism tools and code graders give a similarity score or a pass/fail result, not a skill profile. Learning platforms and resume tools depend on skills that students write about themselves, and on keyword matching. This is not proof of ability. No tool gives the university a summary built from verified student work.

- **Why software and why AI.** Software is needed to store evidence, calculate scores and show dashboards to three types of users. AI is needed because the manual work today is about understanding text, code and images:

Manual work today	Done by AI in SkillTwin
Teacher reads PDFs, code and screenshots to judge skills	Evidence Analysis Agent finds skills and gives evidence scores
Same skill written in different words, such as "Python coding" and "Python development"	Skill Normalization Agent maps them to one ESCO skill
Student reads job advertisements and guesses the needed skills	Job Analysis Agent extracts the required skills
Teacher writes extra practice tasks and marks them	Task Agent and Assessment Agent create and check tasks
Staff study reports to find skill trends	Insight Agent explains the trends in plain language

- **Scope:** The problem is limited to student skill analysis and job-specific skill-gap detection. The system will not perform recruitment, hiring, official grading, attendance management, or university administration.

1.3 Objectives

General objective

Build a web system that turns real student work into a verified skill profile, shows the gap to a target job, and gives personal practice.

AI objective

- Find skills in student work, with evidence.
- Map skills to ESCO.
- Extract skills from job descriptions.
- Create, assess and adapt practice tasks.
- Write insights from backend statistics.

1.4 Scope

In scope	Out of scope
• Lab tasks and submissions • GitHub reading (read only) • Student SkillTwin • Target job and skill gap • Learning path and tasks • Three dashboards • Verified resume	• Course, schedule, attendance • Workshops, enrollment, budget • Running student code • Deciding plagiarism • Exams and certificates as evidence • Automatic academic decisions

AI will	AI will not
• Judge evidence • Map skills to ESCO • Extract job skills • Write tasks and paths • Assess submissions	• Calculate final scores • Claim plagiarism • Claim code was run • Invent data or IDs • Make academic decisions

Boundaries: 3 evidence sources, max 3 task cycles per gap, no management features.

1.5 Stakeholders

Stakeholder	Role	Interest
Students	Submit work, follow learning path	Know skills, close gaps
Teachers	Create lab tasks, view results	See weak skills
University (client)	View aggregate dashboard	Understand skill trends
Administrator	Manage users	Secure, stable system
ESCO (skill standard)	Provides skill names	Consistent naming
University data policy	Rules for student data	Safe data use

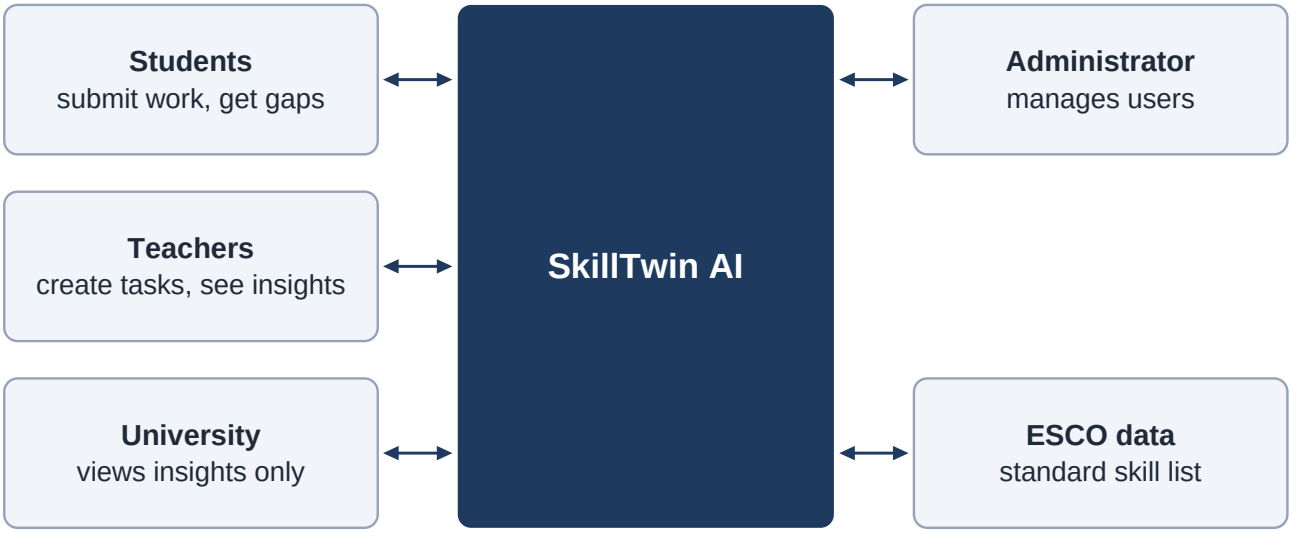


Figure 1: Stakeholders and their interaction with SkillTwin AI.

1.6 Proposed Solution

AI works in the blue steps. The backend stores data and does all calculations.

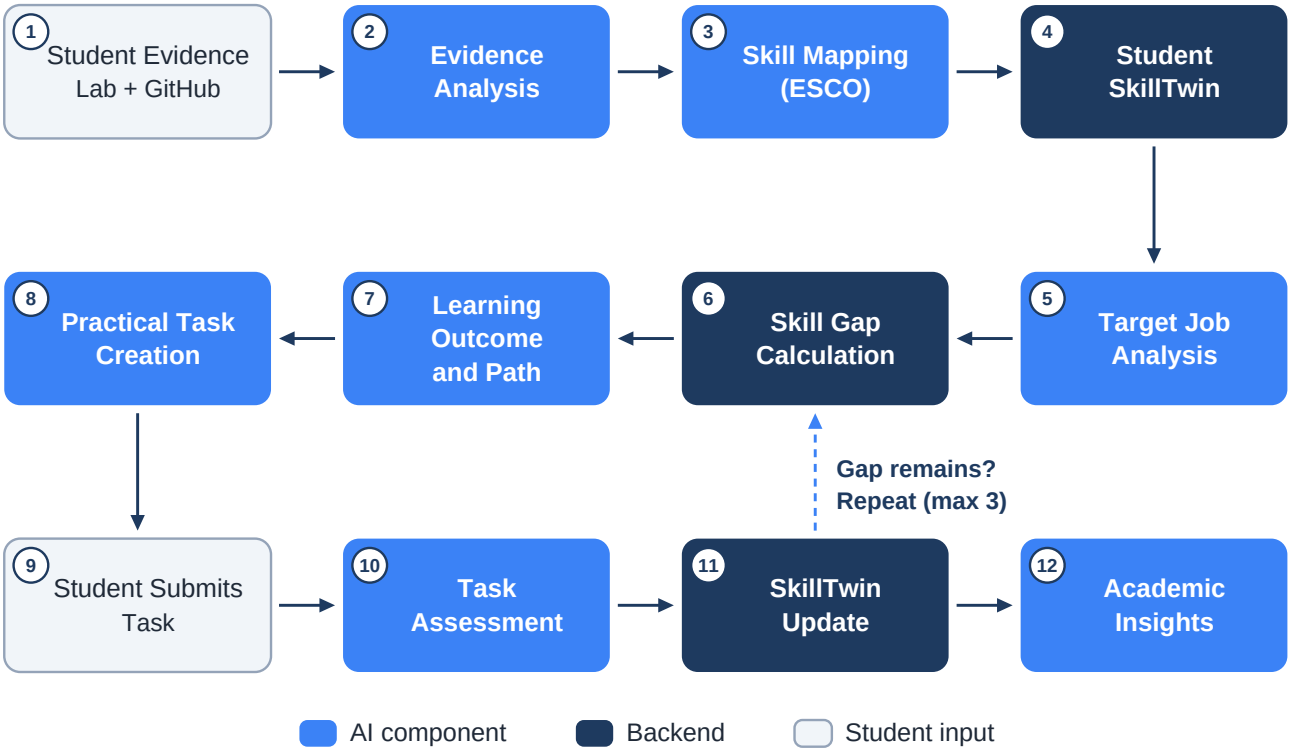


Figure 2: Proposed solution workflow.

Key features

- Lab tasks and project evidence with scores and reasons.
- Target job analysis and skill gap.
- Learning path and practical tasks.
- Teacher and university dashboards.
- Verified resume.

1.7 AI Features

Existing AI features and limits

Existing feature	Limitation
Code auto-graders	Pass or fail only. No skill map.
Similarity checkers	Check copying, not skills.
Learning chatbots	Do not read student work.
Job portals	Use claimed skills, not evidence.

Proposed AI features

Feature	Input	Output	Replaces
1 Evidence Analysis	Task, PDF, GitHub files	Skills, scores, evidence	Teacher reading work
2 Skill Mapping	Skill names	ESCO match	Matching names by hand
3 Job Analysis	Job description	Required skills	Guessing job skills
4 Learning Analysis	Gap, subskills	Outcome, path	Study planning
5 Practical Task	Subskills, last result	Task, rubric	Writing practice tasks
6 Assessment	Task, submission	Scores, evidence	Marking tasks
7 Academic Insight	Backend statistics	Insight text	Writing reports

AI techniques

- **LLM integration:** one prompt and JSON format per agent.
- **AI agents:** 7 agents, one Node.js orchestrator.
- **RAG over own data:** ESCO skills from our database go to the model.
- **Image input:** the LLM reads screenshots.
- **Not in version 1:** own ML model, speech, Bangla AI.

Why AI is essential

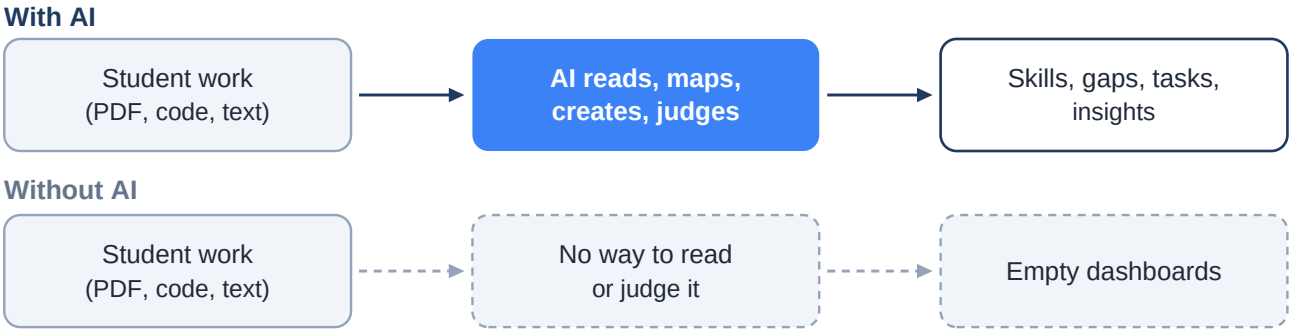


Figure 3: Student work processed with and without AI.

- The backend alone cannot find skills in text and code.
- Without AI there are no scores, requirements, tasks or assessments.
- AI is the core of the product, not an add-on.

2. System Requirements

Functional requirements

ID	Requirement
FR1	The system shall let users register and log in by role.
FR2	The system shall limit each role to its own pages.
FR3	The system shall let teachers create lab tasks with required skills.
FR4	The system shall let students submit PDF, GitHub URL, screenshots and explanation.
FR5	The system shall link each submission to its lab task.
FR6	The system shall let students add GitHub projects.
FR7	The system shall show each student a SkillTwin with score, level, evidence and history.
FR8	The system shall calculate final scores with weights 25 / 25 / 50.
FR9	The system shall let students enter a target job.
FR10	The system shall show required score, gap and "Requirement Met" or "Skill Gap".
FR11	The system shall give a learning outcome, path and task for each gap.
FR12	The system shall let students submit tasks and see the assessment.
FR13	The system shall allow at most 3 task cycles per gap.
FR14	The system shall show teachers common gaps and weak subskills.
FR15	The system shall show the university a dashboard with 8 charts.
FR16	The system shall build resumes from verified data only.
FR17	The system shall keep feedback from changing skill scores.

AI requirements

ID	Requirement
AI1	The system shall find skills in a submission and give a 0-100 evidence score with evidence.
AI2	The system shall give no score when evidence is not enough.
AI3	The system shall tell a skill mention from a skill demonstration.
AI4	The system shall map skills to ESCO or mark them "unmapped".
AI5	The system shall extract required skills from a job description.
AI6	The system shall create a learning path and task from missing subskills.
AI7	The system shall assess a task with rubric scores and confidence.
AI8	The system shall base each next task on the last assessment.
AI9	The system shall write insights only from backend statistics.
AI10	The system shall accept only valid JSON from AI.

Non-functional requirements

ID	Requirement
NFR1	Passwords shall be hashed. The admin code stays on the backend.
NFR2	The system shall show a message while AI works or fails.
NFR3	The system shall work on desktop and mobile browsers.
NFR4	Students shall see only their own data.
NFR5	Every score shall be between 0 and 100.

3. Tools and Technologies

Category	Tools
Languages	JavaScript, HTML, CSS
Frameworks and libraries	React + Vite, Express, Mongoose, Recharts, Multer, pdf-parse, bcrypt, JWT, Zod, Jest
Database	MongoDB
AI models and APIs	LLM API (Claude API) with JSON output and image input
AI data	ESCO skills in MongoDB (demo data first). MongoDB text search. No vector database in version 1.
External API	GitHub REST API (read only)
Dev tools	Git, GitHub, VS Code, Postman

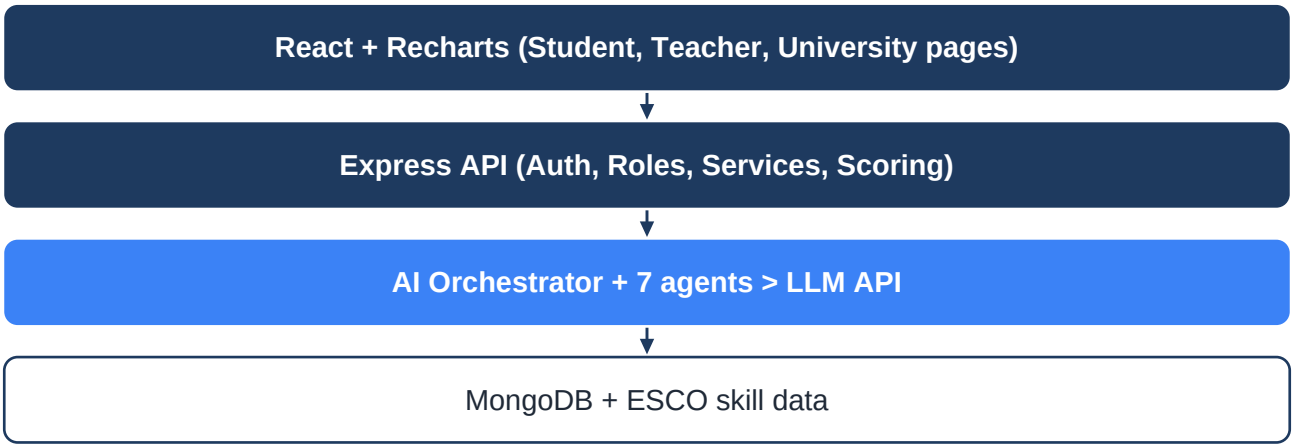


Figure 4: Technology stack layers.

4. Project Timeline and Work Plan

Phase	W1	W2	W3	W4	W5	W6	W7	W8
1. Setup, login, database, ESCO								
2. Lab tasks, submission, PDF								
3. Evidence agents, SkillTwin								
4. Job analysis and skill gap								
5. Learning, task, assessment								
6. University insights, resume								
7. Testing, documents, demo								

Figure 5: Project Gantt Chart. Phase 7 has been updated to take place in W8.

No.	Weeks	Milestone	Deliverables
M1	1	Foundation	Project setup, database models, login and roles, ESCO import, demo data
M2	2	Lab module	Lab task creation, student submission, PDF reading
M3	3-4	Evidence and SkillTwin	Evidence agent, GitHub analysis, skill normalization, score engine, SkillTwin page
M4	5	Job and gap	Job analysis agent, required score, gap calculation, gap page
M5	6	Adaptive learning	Learning, task and assessment agents, orchestrator, 3-cycle loop
M6	7	University and resume	Statistics, charts, insight agent, opportunity planning, resume
M7	8	Final Delivery	Testing, bug fixing, documents, demo practice, final report

4.1 Work Plan: Team Task Division

Area	Faujia	Sabiha	Nipun
Backend	ESCO data, skills APIs, PDF and GitHub readers	Login, roles, lab APIs, scoring, gap, aggregation	Learning, task, feedback, resume APIs
AI agents	Evidence, Skill Mapping, Job Analysis, Orchestrator	Assessment, Academic Insight	Learning Analysis, Practical Task
UI pages	SkillTwin, labs, projects, growth, student skills	Teacher pages, university dashboard, charts	Student dashboard, job, gaps, learning, tasks, results, resume
Testing	AI tests	API and scoring tests	UI and adaptive loop tests
Other	Lead work, proposal, final report	Admin security, demo data	User guide, demo script

- Everyone reviews code and tests their own work.

5. Optional Sections

5.1 Existing System

- Grade systems store marks, not skills.
- Job portals match CV keywords. CVs are self-written.
- No common tool links student work, ESCO and target jobs.

5.2 System Architecture

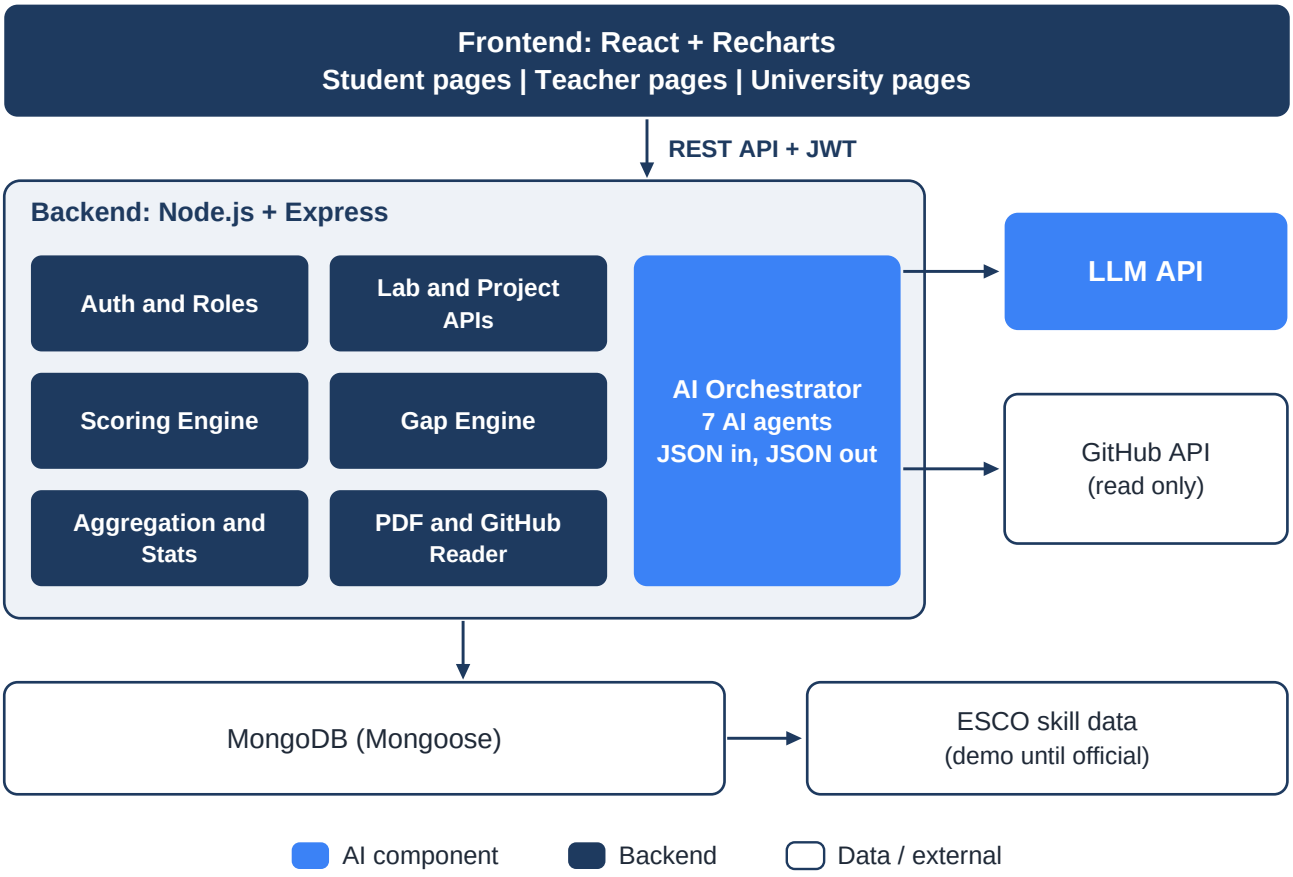


Figure 6: System architecture.

Core data model. The main models and their links are shown below. Other models: User, ProjectEvidence, SkillScoreHistory, Occupation, OccupationSkill, Course, UniversityOpportunity and Resume.

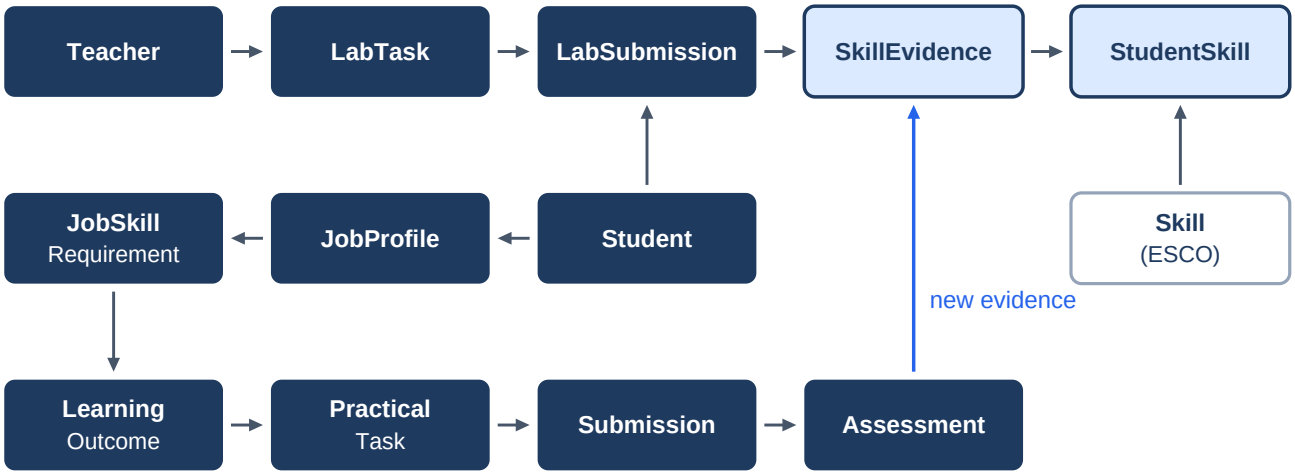


Figure 7: Core data model.

5.3 Ethical, Legal and Social Considerations

- **Privacy:** store only needed data. The university sees aggregates.
- **Fairness:** every score shows evidence and confidence.
- **No plagiarism claims:** originality is only a hint.
- **Transparency:** facts are Observed, Inferred or Unknown.
- **External AI:** send only needed text, never passwords.
- **Human decision:** scores are not grades.

5.4 Future Work

- Load full official ESCO data.
- Add vector search.
- Add Bangla support.
- Pilot with a real class.

6. Summary

SkillTwin AI turns real student work into verified skills, finds gaps for a target job, and creates practice tasks. The backend keeps scores honest. Teachers and the university see insights and decide for themselves. Three members will build it in 8 weeks.